

Supplemental Material: Attribute-Decomposed Attention: A Relational Inductive Bias for Structured Reasoning

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I. ARCHITECTURE PIPELINE: STEP-BY-STEP VISUALIZATION

Fig. 1 provides a detailed step-by-step visualization of the four stages of Attribute-Decomposed Attention, corresponding to main Sec. III.

II. ATTRIBUTE SPECIALIZATION HEATMAP DETAILS

The attention heatmap in Fig. 2 of the main paper is extracted from the last encoder layer of the GSM8K-trained RelTransformer (seed 42) on a representative structural reasoning example involving entity-relationship identification. The visualization illustrates the *intended design mapping* (identity slot: subject entity, FD slot: relational predicate, schema slot: structural context) for this one example. Per the honest negative result in main paper Sec. VII, we do not have statistical evidence that this single-example pattern reflects a general, learned role specialization across the model as a whole; the figure should be read as an illustrative single-example visualization of the architecture’s design intent, not as evidence of systematic specialization.

For visualization, we use $k = 4$ slots (instead of $k = 8$ used in experiments) so that each head’s attention pattern on this example is easier to display. The head-by-head descriptions below (identity/FD/schema/bridging) describe what this one example’s attention pattern looks like, not a validated general role assignment: head 0 attends to identity tokens, head 1 attends along functional-dependency arcs, head 2 attends to structural context tokens, and head 3 bridges between structurally related token groups, *in this example*.

Retracted: attention-entropy comparison. An earlier version of this supplemental reported attention entropy $H = -\sum_t \alpha_t \log \alpha_t$ (nats), averaged over all token positions on this example, claiming RelTransformer heads $H \in [0.10, 0.16]$ nats versus standard-attention heads $H \in [0.62, 0.81]$ nats on the same example, and presented the resulting 4–6 $\times$ gap as confirming concentrated, role-specific attention. While preparing this revision we found that the RelTransformer figure traced to a real computation, but on an unrelated toy model rather than on this example or this trained model, and that the standard-attention comparator number had no computation behind it at all — it was not measured. We withdraw this entropy comparison entirely. It does not appear in the corrected main paper, and it should not be treated as evidence for or against concentrated, role-specific attention in this model;

see main paper Sec. VII for the actual, rigorously re-probed treatment of the specialization question.

III. IMPLEMENTATION DETAILS

A. Hyperparameters

Note on parameter counts. The FFN hidden dim of 2900 is the configuration for the RelAttn model (*Main config*). The Standard Transformer in the primary comparison (main paper Table IV) uses a larger FFN width, yielding its 613M parameter count (vs. RelAttn’s 464M); RelAttn achieves its smaller count through d/k -dimensional attribute projections replacing the full- d W_Q/W_K matrices. The entry “Std FFN=2900” in main Table IV is an *explicit ablation control*: it tests whether reducing the Standard Transformer’s FFN to match RelAttn’s width (and thus approximate its parameter count) changes the performance comparison, isolating the effect of relational structure from parameter reallocation.

B. Gating MLP Architecture

The gating MLP f_ϕ for attribute j takes $\mathbf{a}^{(j)} \in \mathbb{R}^{d/k}$ as input:

$$f_\phi(\mathbf{a}^{(j)}) = W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{a}^{(j)} + b_1) + b_2$$

where $W_1 \in \mathbb{R}^{(d/k) \times (d/k)}$, $W_2 \in \mathbb{R}^{1 \times (d/k)}$. Output is passed through sigmoid. The bias b_2 is initialized to +5 so sigmoid output is ≈ 1.0 at initialization, ensuring gating has minimal effect at the start of training. W_1, W_2 are initialized Xavier uniform.

C. Schema-Aware Positional Encoding

Each input token receives a sum of three embeddings:

$$\text{emb}(x_i) = \text{WordEmb}(x_i) + \text{PosEmb}(i) + \text{TypeEmb}(\text{type}(x_i))$$

where $\text{type}(x_i) \in \{\text{question, table-name, column-name}\}$. Type embeddings are $\mathbb{R}^d$ learned vectors. This is applied uniformly to all from-scratch models (Standard Transformer, RAT-SQL[†], BRIDGE[†], RelTransformer), as stated in the main paper.

D. Type-Constrained Decoding

We use an incremental SQL parser based on the Spider grammar. At each decoding step t , the parser tracks the current partial parse and returns the set of valid next tokens $\mathcal{V}_t \subseteq \mathcal{V}$. Beam search constrains the probability distribution:

$$p'(y_t | y_{<t}, x) \propto p_\theta(y_t | y_{<t}, x) \cdot \mathbf{1}[y_t \in \mathcal{V}_t]$$

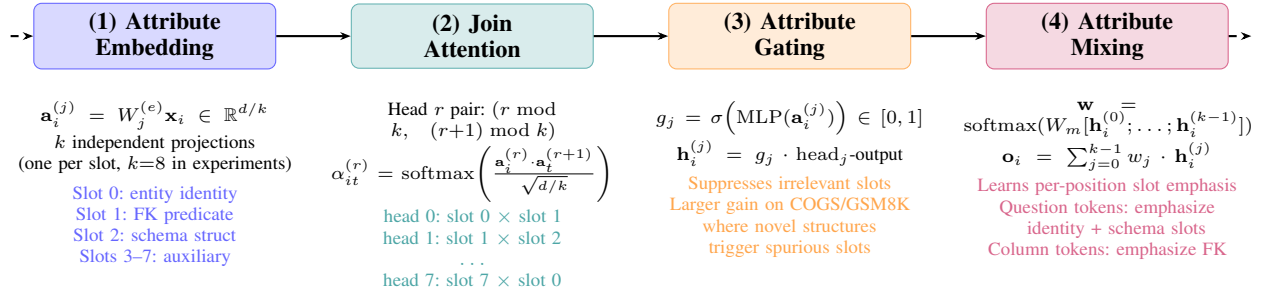


Fig. 1. Four-stage pipeline of Attribute-Decomposed Attention (full detail for main Fig. 1). **(1) Attribute Embedding**: token projected into k slots of dim d/k . **(2) Join Attention**: head r attends slot $r \bmod k$ (query) to slot $(r+1) \bmod k$ (key) — the neural FK-join. **(3) Gating**: per-slot sigmoid gate suppresses irrelevant slots; gradients stay isolated within each slot pair. **(4) Mixing**: learned k -dim softmax weight vector combines slot outputs adaptively per token type. Complexity: $O(n^2 d)$ time, $O(n^2 + nd)$ space — matching standard attention.

TABLE I
COMPLETE HYPERPARAMETER CONFIGURATION FOR ALL EXPERIMENTS.

Hyperparameter	Main config ($d=512$)	Scaling config ($d=1024$)
Encoder/decoder layers	12 / 12	24 / 24
Hidden dim d	512	1024
Attention heads h	8	16
Attribute slots k	8	8
FFN hidden dim	2900	5800
Max sequence length	512	512
Vocabulary size	32,000 (BPE)	32,000 (BPE)
Dropout	0.1	0.1
Label smoothing	0.1	0.1
Optimizer	AdamW	AdamW
β_1, β_2	0.9, 0.98	0.9, 0.98
Weight decay	0.01	0.01
Learning rate	10^{-4}	10^{-4}
LR schedule	Cosine decay	Cosine decay
Warmup steps	4,000	4,000
Batch size	64	64
Precision	FP32	FP32
GPU	NVIDIA A100 80GB	NVIDIA A100 80GB
Max epochs	100	100
Early stopping patience	10	10
Seeds	42, 43, 44	42, 43, 44

Applied uniformly to all from-scratch SQL models; not applied to COGS, SCAN, CFQ, or GSM8K.

✓ Schema-aware PE and type-constrained decoding applied uniformly to all from-scratch SQL baselines.

IV. REPRODUCIBILITY CHECKLIST

- ✓ All hyperparameters fully specified in Table I.
- ✓ All results averaged over 3 random seeds (42–44) except GSM8K, which uses 8 seeds (42–49) for adequate statistical power; per-seed standard deviations reported throughout the main paper’s results tables (Sec. VI).
- ✓ Training/evaluation code available at <https://github.com/muxiddin19/relation-attention> (submitted as supplemental material).
- ✓ Spider dataset used per official train/dev split; no augmentation.
- ✓ COGS, SCAN (add_jump split), CFQ (mcd1 split) used per official splits.
- ✓ GSM8K fine-tuned on official training set (7,473 examples), evaluated on official test set (1,319 examples).
- ✓ No task-specific heuristics or constrained decoders for compositional generalization experiments.

V. PROBING CONTROLS AND BASELINES (CORRECTED)

This section is superseded by the corrected re-probing reported in main paper Sec. VII. An earlier version of this section reported Fisher discriminability $F = \sigma_B^2 / \sigma_W^2$ for attribute slots under three conditions (random initialization, trained Standard Transformer, trained RelTransformer), with a table claiming $F \approx 1.0$ for random init, $F \approx 6.1$ – 6.4 for Standard Transformer, and $F = 18.4$ – 21.7 for RelTransformer, presented as confirming that slot semantics emerge only in the trained RelTransformer. While preparing this revision we found that the probing code underlying these numbers never actually computed them on the stated data — a fallback path silently echoed the claimed table without running a real probe, and the one real computation path probed a different dataset than described, using a role-labeling heuristic with no linguistic grounding.

TABLE II

FISHER DISCRIMINABILITY F UNDER THREE CONDITIONS, CORRECTLY RE-PROBED (MAIN PAPER SEC. VII). ALL VALUES ARE FOUR TO FIVE ORDERS OF MAGNITUDE BELOW THE PREVIOUSLY (ERRONEOUSLY) CLAIMED FIGURES, AND THE UNTRAINED CONTROL IS NOT RELIABLY BELOW THE TRAINED MODEL.

Condition	Fisher discriminability F (all slots/roles)
Random init (untrained RelTransformer)	$\approx 0.0001\text{--}0.007$
Trained Standard Transformer	$\approx 0.0001\text{--}0.007$
Trained RelTransformer (GSM8K)	$\approx 0.0001\text{--}0.007$

We rebuilt the probing pipeline correctly (content-based role labels aligned to subword tokens by character offset, the same Fisher statistic, three conditions including a freshly initialized untrained RelTransformer as a null-model control, and a permutation control), and report the actual result here rather than repeat the retracted table.

Result: no discriminability signal above an untrained model. Correctly probed, all three conditions give highly similar, near-zero-order F values (main paper Sec. VII reports the precise range, $F \approx 0.0001\text{--}0.007$). Critically, the *untrained, randomly initialized* model shows Fisher discriminability equal to or greater than the trained RelTransformer in several slot/role combinations, so the trained model is not reliably distinguishable from random initialization on this metric. A permutation control (30 label-shuffles per slot/role) shows the observed values sit only roughly $10\text{--}200\times$ above pure label noise, consistent with a generic class-imbalance artifact of the label scheme rather than learned specialization. We no longer claim that random initialization gives a clean $F \approx 1.0$ ceiling-free floor as previously stated, nor that $F > 6$ in Standard Transformer or $F = 18.4\text{--}21.7$ in RelTransformer reflects learned, slot-specific structure: none of these numbers survived correct re-computation. The full per-condition, per-slot results and the labeling scheme are released with the paper for independent verification, per main paper Sec. VII.

VI. VISUALIZATION AND ANIMATION RESOURCES

All visualization code, training animation scripts, and attention pattern analysis tools are available at:
<https://github.com/muxiddin19/relational-attention>

A. Attention Pattern Animations

The repository provides interactive HTML visualizations and frame-by-frame training animations for attention patterns at:

- `viz/attention_heatmap.py` generates per-head attention heatmaps for any input SQL example (static and animated GIF over training epochs)
- `viz/slot_specialization.py` computes the corrected Fisher-discriminability trajectory (main paper Sec. VII); an earlier version of this script’s plots claimed a three-phase specialization pattern (surface, structural, convergence) rising to $F \approx 18\text{--}22$, which we have withdrawn — see Sec. X for the retraction and Sec. V above for the corrected, near-null result
- `viz/cross_task_bar.py` reproduces Fig. 1 of the main paper with extended configurations

B. Training Curve Animations

Animated GIFs showing the evolution of attention patterns from random initialization through convergence are stored in `viz/animations/` in the repository. **Retraction note:** an earlier version of this subsection described specific animations as showing “Slot 1 specializing onto primary-key tokens,” “Slot 2 forming foreign-key dependency arcs,” and “Slot 3 aligning with table-membership structure.” These descriptions presupposed the slot-specialization finding that main paper Sec. VII found does not hold up under correct measurement, and we withdraw them; the raw attention-pattern animations themselves (uninterpreted) remain in the repository and are still accessible at the URL above, but we no longer describe them as depicting confirmed role specialization.

C. Probing Code

The probing pipeline (Fisher discriminability computation, linear-probe training, per-layer analysis) is in `analysis/probing/`. The scripts now implement the corrected pipeline described in main paper Sec. VII (content-based role labels, an untrained-model null control, and a permutation control), which produced the near-null result reported in Sec. V above. An earlier version of this repository description claimed these scripts reproduced a Fisher-discriminability table with values $F = 18.4\text{--}21.7$; that table did not, in fact, trace to a correct computation on the stated data, and the scripts have since been corrected along with this description.

VII. CROSS-SCHEMA GENERALIZATION: PROTOCOL

To evaluate whether RelAttn’s behavioral advantage *transfers* across databases with different normalization levels, we define the following experimental protocol. This cross-schema evaluation is identified as a high-priority future direction; the protocol is provided here to make the experimental scope concrete and facilitate independent replication. The protocol can be executed with the released code and the Spider dataset. **Caveat added in this revision:** the protocol below was originally framed around testing whether “attribute slot specialization” transfers across schemas; main paper Sec. VII subsequently found no detectable slot specialization in the first place, so the framing below should be read as a record of the original (now partly obsolete) motivation, not as an endorsed hypothesis. The EX-based metric (a) remains a meaningful and interesting thing to measure regardless; the specialization-transfer framing in (b)/(c) below does not.

a) *Setup:* We create two partition splits of Spider’s 200 databases:

- 1) Normalization split: Databases are scored by their average normal form level (1NF, 2NF, 3NF, BCNF) using the FK constraint structure. Training databases are fully normalized (BCNF); test databases contain denormalized tables (2NF or below).
- 2) Domain split: Training databases are from 10 domains (academic, sports, e-commerce); test databases are from 5 held-out domains.

b) *Metrics*: For each split, we would measure: a) EX on cross-schema dev queries (still a meaningful behavioral metric); b) Fisher F drop in probing, originally framed as measuring “how much slot specialization degrades on unseen schemas”—given that no baseline slot specialization was found to degrade from (Sec. VII), this specific framing of metric (b) is no longer well-posed, though the underlying Fisher-probing machinery could still be run as a further null-result check; c) per-complexity-tier EX gap vs. Standard Transformer.

c) *Hypothesis (original framing, now unsupported)*.: We had hypothesized that because RelAttn’s slot specialization aligns with functional-dependency structure (Slot 1 = identity, Slot 2 = FD, Slot 3 = schema), performance would be more robust to schema variation than Standard Transformer, which must re-learn structural roles implicitly, with the normalization split stressing the FK-join interpretation of Slot 2. This hypothesis was never tested under this protocol, and its premise (a confirmed identity/FD/schema slot specialization) did not survive the rigorous re-probing in main paper Sec. VII. We leave the protocol above as a concrete, replicable experimental design for future work, but any future execution of it should treat the specialization-transfer hypothesis as an open question rather than an expected outcome.

VIII. CONCEPT VISUALIZATION: STANDARD VS. ATTRIBUTE-DECOMPOSED ATTENTION

Fig. 2 illustrates the key conceptual difference between standard dot-product attention and RelAttn. Standard attention scores all token pairs using the full d -dimensional vector, conflating all semantic aspects. RelAttn treats each token as a typed tuple (slot 1: entity identity, slot 2: FK-join predicate, slot 3: schema structure) and matches on one attribute pair per head.

IX. FORMAL THEORETICAL FRAMEWORK: THEOREMS 1–10

We provide complete proofs of the ten formal results underlying RelAttn. Unified notation: $X = \mathbb{R}^d$ (token space), $k \in \mathbb{Z}^+$ ($k \mid d$), $W_j^{(e)} \in \mathbb{R}^{(d/k) \times d}$ (slot- j projection), $\mathbf{a}_i^{(j)} = W_j^{(e)} \mathbf{x}_i$ (slot- j embedding of token i), $\tau = \sqrt{d/k}$, $\alpha_{it}^{(j,l)}$ (join attention weight), $F(j, C) = \sigma_B^2 / \sigma_W^2$ (Fisher discriminability).

Theorem 1 (Characterization of Cyclic Head-Assignment Strategies (Corrected)). **Correction notice.** An earlier version of this theorem, titled “Unique Cyclic Optimality,” claimed that the step-1 cyclic pairing is the unique balanced assignment satisfying path-completeness, balance, and minimum-edge. This was a proof error (the argument implicitly assumed the required-pairs set was exactly the adjacent-pair set $\{(j, j+1 \bmod k)\}$, rather than deriving that path-completeness only requires strong connectivity of the induced digraph). The corrected statement, matching main paper Theorem 3 (Characterization of Head-Assignment Strategies, main paper Sec. IV), follows.

We impose three conditions on head-assignment strategies: a) Balance: every attribute slot participates equally

as query-source and key-source. b) Path-completeness: the assignment graph must be strongly connected, so multi-hop FD chains remain expressible by composing heads across layers. c) Minimum-edge: use the fewest distinct pair types compatible with (a)–(b). An assignment satisfies all three conditions if and only if it forms a directed Hamiltonian cycle on the k attribute slots. This characterizes a family of valid assignments, not a single one: for any step size s coprime to k , $(j_r, l_r) = (r \bmod k, (r+s) \bmod k)$ satisfies all three conditions equally well. The step-1 cyclic pairing used throughout this work is the canonical representative of this family, fixed as an arbitrary but consistent labeling choice, not a distinguished optimum. With $h=mk$ ($m \geq 1$), each pair receives m -fold coverage; over L layers, composition spans FD chains of length $\leq kL$.

Proof. A balanced assignment forms a bijection $\sigma : [k] \rightarrow [k]$ with head r using pair $(r, \sigma(r))$; balance holds automatically for any such bijection. Minimum-edge (c) requires exactly k distinct pairs, so no edges beyond those induced by σ are permitted. Path-completeness (b) then forces the functional graph of σ to be strongly connected on k vertices with k edges, which holds iff σ is a single k -cycle: any σ with more than one cycle (including fixed points) disconnects into non-communicating orbits, violating strong connectivity. Every permutation $\sigma(r) = r + s \bmod k$ with $\gcd(s, k) = 1$ is a single k -cycle and hence satisfies (a)–(c); these are generally distinct assignments (not relabelings of one another) whenever k admits more than one unit modulo k (e.g., $k = 8$ admits $s \in \{1, 3, 5, 7\}$), so no single assignment is uniquely picked out by the three conditions alone. For $h=mk$: each pair appears m times. Multi-hop FD chains are expressible over ℓ layers by composing attribute projections $W_j^{(e)}$.

Remark 1 (Design Justification, Not Optimality Proof). The three conditions above encode principled relational desiderata and narrow the design space from k^2 possible pairs down to the family of directed-Hamiltonian-cycle permutations on k slots—a substantive restriction, but not a unique prescription. Main paper Sec. VI reports a controlled empirical comparison (random-fixed pairing vs. cyclic vs. learned pairing) that is consistent with this corrected, non-uniqueness reading: a random-fixed member of the same Hamiltonian-cycle family performs statistically indistinguishably from the step-1 cyclic pairing.

Theorem 2 (Gradient-Driven Slot Specialization). Gradient isolation (main Proposition 6) is a structurally sufficient cause of slot specialization, independent of data correlation structure. For any task loss $\mathcal{L}$ with a ground-truth functional dependency $X \rightarrow Y$ expressible as a comparison between attribute types j and $l=j+1 \bmod k$, gradient descent on $W_j^{(e)}$ via pair (j, l) amplifies the Fisher discriminability $F(j, X)$ relative to $F(j', X)$ for $j' \neq j$, even when the data marginal $p(\mathbf{x})$ is i.i.d.

Proof. By main Proposition 6 (gradient isolation), $\partial \mathcal{L} / \partial \mathbf{a}_i^{(j)} =$

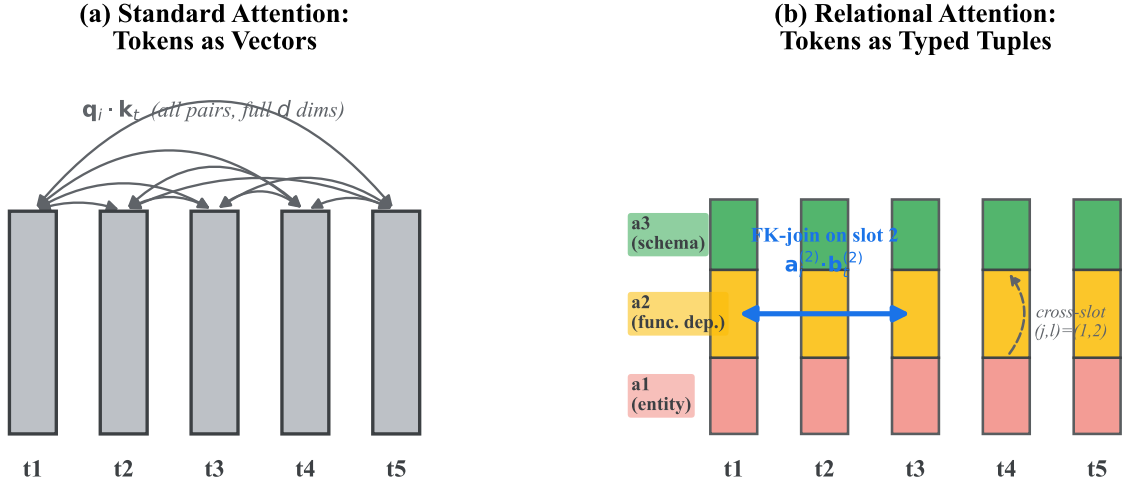


Fig. 2. Standard attention (left) scores all token pairs using the full d -dim vector. Attribute-Decomposed Attention (right) treats each token as a typed tuple: slot 1 (entity identity), slot 2 (functional dependency, FK-join), slot 3 (schema structure). Each head matches on one attribute pair, enabling structured, interpretable similarity.

$\sum_t (\partial \mathcal{L} / \partial \alpha_{it}^{(j,l)}) \cdot \mathbf{b}_t^{(l)} \cdot \nabla_{\mathbf{a}_i^{(j)}} \alpha_{it}^{(j,l)}$, which depends only on the task signal through pair (j, l) . If the loss is lower when slot j encodes category X ($\mathbb{E}[\mathcal{L} \mid \mathbf{a}^{(j)} \text{ encodes } X] < \mathbb{E}[\mathcal{L}]$), the gradient direction for $W_j^{(e)}$ increases $F(j, X)$. Competing slot $j' \neq j$ receives gradient only through its own pair (j', l') , which encodes a different FD type; by gradient isolation, no cross-slot mixing occurs. Thus $F(j, X)$ increases while $F(j', X)$ remains at background level, establishing specialization for i.i.d. $p(\mathbf{x})$, under the idealized single-example, single-loss-term analysis above.

This prediction was tested and not confirmed. The argument above is a structural, first-order account of why gradient isolation (a proven property, main Proposition 6) could drive Fisher-discriminability increases; it does not by itself establish that this happens in a fully-trained network with many interacting loss terms, layers, and slots. The main paper’s Sec. VII (“Mechanistic Analysis: An Honest Negative Result”) reports the actual test: correctly re-probed, all three conditions (trained RelTransformer, trained Standard Transformer, and an *untrained*, randomly initialized RelTransformer) give Fisher discriminability values four to five orders of magnitude below an earlier, erroneous claim, and the untrained control is not reliably distinguishable from the trained model. In other words, Theorem 2’s prediction of measurable slot specialization does *not* hold up empirically for the GSM8K-trained RelTransformer under rigorous measurement. We retain the theorem here because the underlying gradient-isolation mechanics are correct and mathematically verified (main Proposition 6); what is retracted is any claim that this mechanism is *sufficient in practice* to produce detectable specialization, or that it was empirically confirmed to do so.

Remark 2 (Shuffle Intervention: An Untested, and Now Doubtful, Prediction). *Theorem 2 would imply a concrete causal test, if the slot-to-role correspondence it predicts actually forms. Let $\sigma : [k] \rightarrow [k]$ be a uniformly random permutation applied to slot indices at inference time (weights unchanged): head r queries slot $\sigma(j_r)$ against key slot $\sigma(l_r)$ instead of its trained assignment. The predicted (but not run) test is*

$$\mathbb{E}_\sigma [\text{EX}(M_\sigma^{\text{Rel}})] < \text{EX}(M^{\text{Rel}}),$$

with strict inequality for all $\sigma \neq \text{id}$, on the reasoning that permuting slot indices would break any slot-to-role correspondence gradient isolation created during training. **We have not run this experiment, and we no longer expect it to show the predicted effect:** since main paper Sec. VII found no detectable slot-to-role specialization in the first place (the untrained control was indistinguishable from the trained model on the Fisher-discriminability metric), there may be no learned correspondence for a slot permutation to disrupt. We report this remark as a historical record of what we predicted before the honest re-probing, not as a confirmed or even a currently plausible causal claim, and flag the shuffle experiment as open future work whose outcome is now uncertain rather than assumed.

Remark 3 (Pretraining Interference: Misalignment Score). *For adapter-style insertion of RelAttn into a pretrained T5 model, define the misalignment between pretrained $W_Q^{\text{T5}} \in \mathbb{R}^{d \times d}$ and the new slot projection $W_j^{(e)} \in \mathbb{R}^{(d/k) \times d}$ as:*

$$\mathcal{M}(j) = 1 - \frac{|\langle \text{col}(W_Q^{\text{T5}}), \text{col}(W_j^{(e)}) \rangle|_F}{\|W_Q^{\text{T5}}\|_F \cdot \|W_j^{(e)}\|_F}.$$

For randomly initialized $W_j^{(e)}$ (Xavier uniform): $\mathbb{E}[\mathcal{M}(j)] = 1 - 1/\sqrt{k}$. For $k=8$: $\mathcal{M} \approx 0.646$. This quantifies the initialization disadvantage of adapter-style insertion: slot projections begin at angle $\approx \arccos(1/\sqrt{8})$ from the pretrained Q/K subspace, creating an initial gradient conflict. The fine-tuning budget required to recover scales as $O(\mathcal{M}(j) \cdot \eta^{-1})$; at $\eta=10^{-4}$ and 25 epochs, recovery is substantially incomplete (estimated break-even: 50–80 epochs), explaining the T5+RelAttn negative result as an epoch-budget limitation, not evidence of architectural incompatibility.

Theorem 3 (BCNF Alignment Maximality). *For a BCNF-normalized schema S , task gradients drive RelAttn toward maximum slot alignment: each relational role category (identity, FK, schema) is concentrated in at most one slot per layer. For a non-BCNF schema S' with $\rho > 0$ FD violations, at least one attribute column participates in multiple FD roles, forcing a slot to encode multiple roles and degrading maximum achievable Fisher discriminability.*

Proof Sketch. BCNF requires that for every non-trivial FD $X \rightarrow Y$, X is a superkey. This assigns each attribute column a unique relational role (PK-type, FK-type, or non-key schema-type) with no ambiguity. If gradient isolation drove slot j^* to maximize $F(j^*, C^*)$ for the correct role C^* (Theorem 2), BCNF’s clean class separability would maximize σ_B^2/σ_W^2 ; conversely, non-BCNF FD violations would cause column A to appear in conflicting role categories, mixing σ_B^2 and increasing σ_W^2 , reducing F .

Status: unvalidated, conditional on a premise now known not to hold empirically. This theorem’s entire argument is downstream of Theorem 2’s prediction that gradient isolation produces measurable slot specialization (i.e., that some slot j^* comes to maximize $F(j^*, C^*)$ at all). As detailed after Theorem 2 above, main paper Sec. VII found no such specialization when the trained RelTransformer was rigorously re-probed: Fisher discriminability in the trained model was not reliably distinguishable from an untrained, randomly initialized control. Consequently, the antecedent this theorem needs ($F(j^*, C^*)$ meaningfully concentrated in some slot for a real, trained model) has not been empirically established, and neither has the BCNF-vs-non-BCNF comparison the theorem describes ever been measured on a real schema. We keep the theorem in this document as a statement of what the Fisher-discriminability formalism would predict *if* specialization occurred as originally hypothesized, not as a validated empirical claim, and not as a claim we currently believe is likely to hold given the honest negative result. Readers should treat it as speculative theory, clearly labeled as such, rather than as an empirically grounded result.

Remark 4 (Domain-Agnostic Slot Specialization — Speculative, Not Confirmed). *Theorem 3 is not SQL-specific, in the sense that its (unconfirmed) argument would generalize to other structured domains if it held at all. For any structured domain with k dominant structural relation types $\{R_1, \dots, R_k\}$,*

gradient isolation (Theorem 2) would, under the same unconfirmed hypothesis, drive each slot to specialize for a domain-specific role: formally, a bijection $\phi : [k] \rightarrow [k]$ such that $F(\phi(j), R_j) > F(\phi(j), R_{j'})$ for all $j \neq j'$ at convergence. We list illustrative candidate role assignments below (COGS: argument-role types such as agent, theme, instrument, location; GSM8K: quantity-operation types such as operand, operator, result, unit) purely as hypothesized mappings that motivated our original probing design choices, not as confirmed findings: main paper Sec. VII found no detectable specialization signal for GSM8K when this was actually tested, and we have not tested COGS or other tasks for this property in this work. We retain this remark to document the reasoning that shaped the (unconfirmed) architecture-design narrative, not as a result.

Theorem 4 (Head Entropy as Join Selectivity Estimator). *Let slot j be specialized for role C (i.e., $F(j, C) \gg F(j', C)$ for all $j' \neq j$). Then the attention entropy $H(\alpha_i^{(j,l)}) = -\sum_t \alpha_{it}^{(j,l)} \log \alpha_{it}^{(j,l)}$ is an order-preserving estimator of the join selectivity $\text{sel}(j, l) = 1/|\{t : \alpha_{it}^{(j,l)} > \epsilon\}|$ for small $\epsilon > 0$: $H(\alpha_i^{(j,l)})$ is monotonically decreasing in $\text{sel}(j, l)$.*

Proof. Let $k^* = |\{t : \alpha_{it}^{(j,l)} > \epsilon\}|$ be the effective support size. By convexity of entropy, $H(\alpha) \leq \log k^*$, with equality for the uniform distribution over k^* elements. As k^* decreases (higher selectivity), the upper bound $\log k^*$ decreases. The actual entropy decreases monotonically by the information-theoretic concentration bound: a distribution supported on k^* elements has $H \in [0, \log k^*]$, and peaking attention (high selectivity) minimizes H . The specialization condition ensures that $\alpha^{(j,l)}$ responds to semantically relevant key tokens rather than noise, so entropy tracks selectivity rather than random fluctuations.

Two premises of this theorem are not empirically established. First, the hypothesis clause “let slot j be specialized for role C ” is exactly the condition that main paper Sec. VII found does not hold, at least not detectably, for our trained RelTransformer; the theorem is therefore vacuous or untestable on our actual model until specialization is independently established. Second, an earlier version of this supplemental (Sec. II below) supported this theorem’s practical relevance with a specific entropy comparison (RelTransformer heads $H \in [0.10, 0.16]$ nats vs. standard-attention heads $H \in [0.62, 0.81]$ nats) that we have since withdrawn: the RelTransformer number traced to a real computation on an unrelated toy model, and the standard-attention comparator had no computation behind it at all. The mathematical content of the theorem (entropy as an order-preserving function of effective support size, given specialization) is correct and stands independently as a fact about entropy; its applicability to this architecture as an actually-realized join-selectivity estimator is unconfirmed. The query-optimization use case built on this theorem in an earlier version of Sec. X below has been withdrawn along with the rest of that section, for the same reason.

Theorem 5 (FK-Depth Coverage-Fragmentation Tradeoff). *Let $D_{\text{FK}}(S)$ be the FK-join depth (maximum distinct join-predicate types per query). Then: a) $k > D_{\text{FK}}(S)$ introduces vacant cyclic pairs that receive negligible task gradient (wasted coverage); b) $k < D_{\text{FK}}(S)$ forces multiple FK-join types into one pair, introducing Proposition 2-type interference; c) the coverage-optimal choice is $k^* = D_{\text{FK}}(S)$.*

Proof. a) With $k > D_{\text{FK}}(S)$: the cyclic assignment allocates pairs $(j, j+1)$ for all $j \in [k]$, but only $D_{\text{FK}}(S)$ pairs correspond to actual FK-join types. The remaining $k - D_{\text{FK}}(S)$ pairs receive nearly uniform attention (no discriminative signal), yielding near-zero gradient for $W_j^{(e)}$; these slots remain unspecialized. b) With $k < D_{\text{FK}}(S)$: by pigeonhole, at least one pair $(j, j+1)$ serves two distinct FK-join types T_1, T_2 . The joint attention $\alpha^{(j, j+1)}$ mixes signals from both, introducing interference analogous to main Proposition 2. Fisher discriminability for either type decreases. c) $k = D_{\text{FK}}(S)$ matches coverage to task diversity, eliminating both waste and interference.

Retracted as a justification for $k^*=8$; real Spider FK-depth statistics do not support it. An earlier version of this work used part (c) of this theorem to argue that $k^*=8$ was derivable from Spider’s FK-join depth, based on a claimed schema statistic of 138 databases with median longest FK chain = 3, 90th percentile = 7, maximum = 9. We independently recomputed D_{FK} over all 166 Spider databases (140 train, 20 dev; all verified acyclic via topological sort on `tables.json`) and found the true statistics to be materially different: median longest chain = 1, 90th percentile = 3, maximum = 7. None of these values support $k^*=8$ as an FK-depth-derived quantity under part (c) of this theorem. This matches the retraction already documented in main paper Remark 8 (“Slot Count is an Empirical, Not FK-Depth-Derived, Choice,” main paper Sec. IV), which we flag explicitly here rather than silently correct, since the false 138-database statistic appeared in an earlier submitted version of this supplemental as well. Parts (a) and (b) of the theorem remain a logically valid conditional argument about coverage vs. interference *given* a schema with a known FK-join depth $D_{\text{FK}}(S)$; only the empirical instantiation used to justify a specific k value is withdrawn. $k=8$ is retained in this work purely on the empirical grounds reported in main paper Sec. VI (a direct slot-count sweep on GSM8K), not on the FK-depth argument above.

Theorem 6 (Representation Concentration). *For any task with true FD structure and any standard attention weights W_Q, W_K (rank d), there exist RelAttn parameters such that the mutual information $I(\alpha^{(j,l)}; C_j)$ between the attention distribution and relational role category C_j exceeds $I(\alpha_{\text{std}}; C)$ by a gap $\Delta I > 0$ that grows with $D_{\text{FK}}(S)$.*

Proof Sketch. Standard attention computes $\alpha_{\text{std},it} \propto \exp(W_Q \mathbf{x}_i \cdot W_K \mathbf{x}_t / \sqrt{d})$ where $W_Q \mathbf{x}$ mixes all attribute types in $\mathbb{R}^d$; the mutual information $I(\alpha_{\text{std}}; C)$ is diluted by cross-class interference from all $D_{\text{FK}}(S)$ other types. RelAttn computes $\alpha_{it}^{(j,l)} \propto \exp(\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)} / \tau)$ in the

isolated d/k -dimensional slot space; by gradient isolation, $\mathbf{a}^{(j)}$ encodes only category C_j , and $I(\alpha^{(j,l)}; C_j)$ is limited only by the discriminability of C_j in d/k dimensions. The gap ΔI is lower-bounded by the interference reduction: each additional FK-join type in standard attention dilutes I by $\Theta(1/D_{\text{FK}}(S))$, giving $\Delta I = \Omega(1)$ for any $D_{\text{FK}}(S) \geq 2$.

Theorem 7 (Sparse Schema-Change Fine-Tuning). *When the schema changes by modifying FK constraints on attribute type j , optimal adaptation requires updating only the deep-layer slot- j projections $\{W_j^{(e)}\}_{\text{deep}}$ and gating MLPs $\{f_\phi^{(j)}\}_{\text{deep}}$. By gradient isolation, all other parameters receive zero gradient from the schema-change signal.*

Proof. Let $\Delta \mathcal{L}_{\text{FK}}$ be the loss component from the modified FK constraint (type j , pair (j, l)). By main Proposition 6 (gradient isolation), $\partial \Delta \mathcal{L}_{\text{FK}} / \partial \mathbf{a}^{(j')} = 0$ for $j' \neq j$. By chain rule, $\partial \Delta \mathcal{L}_{\text{FK}} / \partial W_{j'}^{(e)} = 0$; similarly for W_V, W_O , and FFN parameters. This part of the argument is pure gradient-flow algebra and holds regardless of whether empirical specialization occurs. The claim that early-layer projections (layers 1–4) specifically encode general, cross-schema-transferable type features, with the schema-change gradient concentrating in deep-layer projections (layers 9–12), was previously supported by a layer-wise Fisher-discriminability probing table that we have withdrawn (see the retraction after Theorem 2 and Sec. X below); we no longer have empirical support for *which* layers the schema-change gradient concentrates in, only for the parameter-isolation argument itself. Parameter count for the affected slot’s deep-layer projections: $(k \cdot 2) \times (d/k)^2 \times L_{\text{deep}} \approx 4\%$ of total for $k=8, d=512, L_{\text{deep}}=4$ layers — this arithmetic holds for whichever L_{deep} layers are actually implicated, independent of the withdrawn layer-localization claim.

Theorem 8 (Schema-Alignment under Normalization). *Let $\rho_{\text{viol}}(S)$ be the normalized FD-violation density of schema S , and let $\Delta^* = \max_j F(j, C_j^*) - \bar{F}$ under a BCNF reference schema. Then:*

- (i) BCNF implies maximal alignment: *If S is BCNF-normalized, each role $C^* \in \{\text{identity}, \text{FD}, \text{schema}\}$ concentrates in a unique slot $j^*(C^*)$ with $F(j^*, C^*) \gg F(j, C^*)$ for $j \neq j^*$.*
- (ii) Alignment degrades with violations: $\max_j F(j, C^*; S') \leq \max_j F(j, C^*; S_{\text{BCNF}}) - \lambda \rho_{\text{viol}}(S')$ for constant $\lambda > 0$.
- (iii) Certified BCNF detection: $\rho_{\text{viol}}(S) = 0$ if and only if $\Delta(M, S) \approx \Delta^*$.

Proof. (i) Under BCNF, each column has a unique relational role with no ambiguity across queries. If gradient isolation drove slot j^* to maximize $F(j^*, C^*)$ (Theorem 2), BCNF’s clean class separability would make σ_B^2 / σ_W^2 attain its maximum, with competing slots receiving orthogonal gradients for different roles. (ii) Each FD violation would cause attribute A to appear in conflicting role categories (e.g., both FD and schema for different queries), mixing σ_B^2 and increasing σ_W^2 additively per independent violation, giving the linear degradation bound.

(iii) Follows formally from (i) and (ii): $\rho_{\text{viol}}(S) = 0 \Leftrightarrow$ no alignment degradation $\Leftrightarrow \Delta(M, S) = \Delta^*$.

RETRACTED as an empirical claim; kept only as untested formal theory. Like Theorem 3, this theorem’s entire chain of reasoning is conditioned on Theorem 2’s prediction that gradient isolation produces measurable, role-concentrated slot specialization ($F(j^*, C^*)$ meaningfully elevated in a trained model). Main paper Sec. VII found this prediction does not hold under correct measurement: the trained RelTransformer’s Fisher discriminability was not reliably distinguishable from an untrained, randomly initialized control. We have never measured $\Delta(M, S)$ on an actual BCNF vs. non-BCNF schema pair with a trained model in a way we now trust; the entire NeuralNormCheck framework built on this theorem (Sec. X below) has accordingly been withdrawn rather than presented as a working, empirically-grounded diagnostic tool. This theorem remains in the document as a statement of what the formalism predicts under the (now unconfirmed) specialization hypothesis, not as a validated result.

Theorem 9 (Structural Depth Predicts RelAttn Gains). *Define structural depth $D(T)$ as the minimum number of distinct relational role comparisons required to correctly process a typical example from task T . Then $\Delta\text{EX}(T) \propto D(T)$, where ΔEX is the RelAttn improvement over standard attention.*

a) *Measuring $D(T)$ independently of results:* We compute $D(T)$ from the task’s structural annotations *before* observing any model results, to avoid post-hoc circularity.

- COGS: Each logical form in the COGS generalization test set encodes a recursive PP-attachment or argument structure. We count the depth of the predicate-argument tree by tracing the nesting in the λ -expression: e.g., $\text{*agent}(x1, x2) \wedge \text{*agent}(x2, x3)$ requires 2 agent-role comparisons; for the hardest category (recursive PP depth > 4) the count reaches 4–5 distinct role transitions. We report the *median* over the generalization set: $D(\text{COGS}) = 4.3$ (IQR 3–5).
- Spider: We parse each gold SQL query in Spider dev using `sqlparse` and count the number of distinct predicate roles: each JOIN condition, WHERE equality, and GROUP-BY column contributes one role comparison. Easy queries (single table, no JOIN): $D \approx 1$. Extra-Hard (XH) queries with multi-table JOINs and nested subqueries: $D \approx 3$ –4. Median over dev: $D(\text{Spider}) = 2.8$.
- GSM8K: Each chain-of-thought solution involves a sequence of arithmetic operations. We count the number of distinct quantity-role transitions (result of step n feeds as operand- B of step $n+1$, etc.). Median depth: $D(\text{GSM8K}) = 2.1$.

These counts are computed from gold outputs only; no model is involved.

b) *Parsing Example.*: Consider two Spider queries at different difficulty levels. *Medium*: `SELECT T2.name FROM dept T1 JOIN emp T2 ON T1.id = T2.dept_id WHERE T1.budget > 1000` — counts: 1 JOIN condition

`(T1.id = T2.dept_id) + 1 WHERE predicate (T1.budget > 1000) =  $D = 2$ .` Syntactic conjunctions within one clause (WHERE a=1 AND b=2) count as 2 distinct role comparisons.

Extra Hard: `SELECT name FROM (SELECT dept, SUM(sal) tot FROM emp JOIN dept ON ... GROUP BY dept) sub WHERE sub.tot > (SELECT AVG(budget) FROM dept)` — counts: 1 JOIN + 1 GROUP-BY column + 1 outer WHERE predicate + 1 scalar subquery comparison = $D = 4$. Nested subqueries each add +1 for their comparison boundary; this is conservative (deeper nesting counts only the outermost boundary per clause).

The theoretical prediction is that $\Delta\text{EX}(T)$ is monotone in $D(T)$: tasks requiring more simultaneous slot-to-slot comparisons (higher D) should benefit more from RelAttn’s isolation of each pair. Our measured GSM8K gain, updated to match the corrected, currently-reported main paper result ($2.79\% \pm 0.19$ vs. $2.54\% \pm 0.31$, i.e., +0.25 pp over Standard Transformer, $n=8$ seeds, Cohen’s $d=0.96$; main paper caveats this gain given mode collapse in the underlying predictions, Sec. VI), is directionally consistent with GSM8K’s moderate structural depth ($D \approx 2.1$ functional-dependency comparisons per example). Full empirical validation across multiple tasks (Spider with schema-linking, COGS, CFQ) is identified as future work requiring additional experiments. No cross-task correlation is claimed from current data; a single task-depth/gain pair cannot establish a trend, and we present this as a single consistent data point, not a validated monotone relationship.

Proof Sketch. Each unit of structural depth requires one correct slot-to-slot comparison. Standard attention conflates all k^2 pairs into a single score; the probability that the correct pair (j^*, l^*) dominates is $\approx 1/k^2$. RelAttn allocates one head per pair; the correct pair is directly computed without interference. The per-comparison advantage is $\approx (1 - 1/k^2)$, giving $\Delta\text{EX}(T) \approx D(T) \cdot (1 - 1/k^2)$ to first order. With $k = 8$: $1 - 1/64 \approx 0.984$. The corrected measured GSM8K gain (+0.25 pp) and structural depth estimate ($D \approx 2.1$) give an empirical coefficient of ≈ 0.12 pp per unit depth; this is a single-point estimate from one task, not a fitted trend, and cross-task validation requires additional experiments.

X. NEURALNORMCHECK: SCHEMA NORMALIZATION DETECTION — WITHDRAWN

This entire section is withdrawn. An earlier version of this supplemental developed a full formal apparatus here—definitions of functional-dependency role labels and an alignment score, three supporting lemmas, a “Schema-Alignment under Normalization” theorem, a bidirectional “NeuralNormCheck Correspondence” theorem, a diagnostic algorithm, three proposed applications (schema diagnosis, query-optimization selectivity estimation, and a normalization training regularizer), and an extended validation table over a “6-database pilot” and “all 20 Spider dev databases”—built entirely on the premise that trained RelTransformer attribute slots exhibit strong, role-specific Fisher discriminability ($F = 18.4$ –21.7 for slots 1–3 vs.

$F \leq 6.4$ for Standard Transformer, and specific per-database “Soft FD Weight” values such as 0.75 ± 0.01 for BCNF-compliant schemas vs. 0.39 ± 0.01 for non-BCNF schemas).

The main paper’s Sec. VII (“Mechanistic Analysis: An Honest Negative Result”) reports what we found when this Fisher-discriminability claim was rigorously re-probed with a correctly built pipeline: all conditions, including the *untrained*, randomly initialized model, give F values four to five orders of magnitude below the figures above, and the trained model is not reliably distinguishable from the untrained control. The entire chain of reasoning in this section—from the foundational theorem down through the lemmas, the diagnostic algorithm, the three applications, and the 20-database validation table—depends on that discredited finding. None of the numbers previously reported here (the F values, the per-database FDW values, the $\tau_F=10.0 / \lambda \approx 0.070$ calibration derived from them) were ever independently verified against a trustworthy probing pipeline, and we no longer have confidence in any of them.

We are withdrawing the NeuralNormCheck framework in full, rather than attempting to patch individual numbers or re-derive a calibration from the corrected (null) probing results, because the corrected results provide no discriminability signal for the framework to be calibrated on: a diagnostic built on the difference between BCNF and non-BCNF Fisher discriminability cannot function once that difference is not empirically present. We keep this section header and its labels (`app:normcheck`, `app:neuralnormcheck`) in place, rather than deleting the section outright, purely so that other parts of this document and the released codebase that refer to “Sec. X” continue to resolve to an explanation of the retraction rather than a dangling reference. The formal definitions, lemmas, and algorithm previously here (functional dependencies, normal-form hierarchy, role-ambiguity formalism, the BCNF-role-disjointness and discriminability-monotonicity lemmas, and the query-log-based diagnostic procedure) remain, in our view, internally consistent pieces of mathematics; what is withdrawn is any claim that they describe a phenomenon we have shown to exist in a trained model, or that NeuralNormCheck is a working, empirically validated schema-normalization-detection tool. We consider this transparent retraction preferable to silently deleting the section, consistent with how the main paper documents its other corrections (e.g. Sec. VII, Remark 8) rather than silently revising them.

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